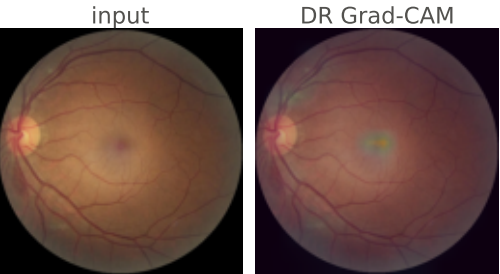


Five-disease retinal screening on low-cost edge hardware

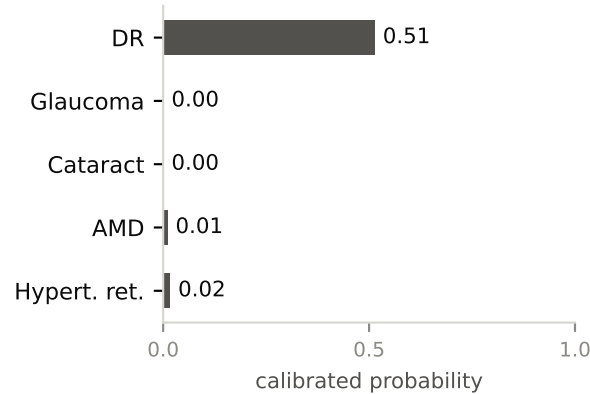
1 One fundus photo



Held-out test image, moderate NPDR (seeded random draw)

2 MobileNetV3-small, 512 px, FP32

6 MB · ODIR-5K · Platt-calibrated outputs



Test AUC (3 seeds): mean 0.846 · DR 0.78
Deployed model*: referable DR AUC 0.87

3a Intel N100 mini PC

Rs 37,500

Test AUC vs workstation	identical
Model latency	8.3 ms
+ Grad-CAM, 5 labels	+4.9 ms
Peak memory	222 MB
Energy (iGPU)	64 mj / image

3b Raspberry Pi 3B

Rs 3,500

Test AUC vs workstation	identical
Model latency	0.51 s
+ Grad-CAM, 5 labels	+146 ms
Peak memory	65 MB of 870 MB
Clock (under-powered)	600 MHz

512 px vs 224 px: DR AUC +0.073 (95% CI 0.047-0.099) Naive INT8: lower AUC, broken calibration, little speed-up → deploy FP32 Platt scaling: DR ECE 0.163 → 0.027

*Final model, median-validation seed. Pi throttled by under-voltage; latencies ≈2× pessimistic. Grad-CAM does not localise microaneurysms.